

GenomeGuard Project Report

Overview

GenomeGuard is a bioinformatics pipeline developed to evaluate the taxonomic integrity of sequencing samples using Centrifuge classifications. The project evolved from simple taxonomic summaries into a multi-stage evidence framework.

Objectives

- Build an automated pipeline from FASTQ to interpretation.
- Quantify dominant and secondary taxonomic signals.
- Avoid over-interpreting secondary classifications as contamination.
- Produce reproducible reports.

Development Timeline

v0.1

- Read classification summary.
- Classification rate.
- Top taxa.

v0.2

- Added evidence metrics including dominance and secondary burden.

v0.3

- Introduced evidence-weighted species table using read counts and unique reads.

v0.4 / v0.4.2

- Robust parsing of Centrifuge reports.
- Normalized species evidence.

v0.5

- Sample integrity engine.
- Dominance categories and QC interpretation.

v0.6

- Ranked secondary signals for investigation.

v0.7

- Read-level validation using Centrifuge scores.

v0.8

- Cross-evidence consistency integrating abundance and score metrics.

v0.9

- Taxonomic conflict analysis grouping related taxa (e.g. *Escherichia/Shigella*).

v0.10

- Discriminative evidence using score-gap analysis.
- Supported / Moderate / Ambiguous evidence classes.

GenomeGuard v1.1

Complete automated pipeline:

1. Input paired FASTQ files.
2. Run Centrifuge.
3. Generate classification and report files.
4. Compute GenomeGuard metrics.
5. Produce TSV summary, HTML report, and log.

Workflow

SRA -> prefetch -> fasterq-dump -> FastQC -> GenomeGuard -> Centrifuge -> Evidence analysis
-> Reports

Key Metrics

- Classification rate
- Unclassified fraction
- Dominant taxon
- Secondary burden
- Taxonomic complexity
- Read-level evidence
- Cross-evidence consistency
- Score-gap discrimination

Major Achievement

GenomeGuard became an end-to-end pipeline executable with:

`./genomeguard.sh sample_1.fastq sample_2.fastq`

Outputs:

- classification.tsv
- report.tsv
- genomeguard_summary.tsv
- genomeguard_report.html
- genomeguard.log

Scientific Principle

GenomeGuard reports computational evidence only. Secondary taxonomic assignments are not interpreted as proof of contamination, co-infection, or mixed biological origin without additional biological validation.

Future Work

- Single-end support
- Multi-sample batch mode
- Interactive HTML dashboard
- Krona plots
- Multi-classifier support
- Containerized release